

Revisiting the calculation of performance margins in monitoring-enabled optical networks

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In the context of future intelligent optical networks, dedicated learning techniques can be employed to monitor physical system parameters with a guaranteed accuracy. In this work, we investigate a method that establishes the link between input parameter uncertainties and the overall performance uncertainty. To this end, neglecting stochastic effects and focusing on the input parameters of a simplified Gaussian noise model version, we employ uncertainty propagation to evaluate the overall performance uncertainty from input parameter uncertainties, and we propose a simple way to link performance uncertainty to margins. With this method, as opposed to direct performance monitoring, it is possible to trace, in a predictable way, the path from the cause (input parameter uncertainties) to the effect (performance uncertainty) and to the additional network-level consequences (performance margins). We briefly review methods used in the literature to set margins in classical systems, and we show how all methods can be unified by means of the *correlation* between input parameters. By quantifying the impact of input parameter correlations, we further discuss the margins that can be saved if input parameters are partially correlated or uncorrelated, compared to a scenario in which parameters are fully correlated. We finally illustrate the separate impact of each parameter on performance uncertainty, and we briefly discuss their order of importance as a function of the system operating regime and propagated distance. © 2019 Optical Society of America

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1. INTRODUCTION

System performance can be analyzed in terms of several different quality of transmission (QoT) metrics before forward error correction (FEC), such as the bit error ratio (BER), the signal-to-noise ratio (SNR), mutual information, etc. In all cases, the word “margin” customarily designates the excess of the measured “on field” system performance compared to the performance required to guarantee error-free operation after FEC. Today, optical networks are commonly designed with a high level of performance margins to ensure that service level agreements (SLAs) are met throughout the system lifetimes. Leveraging unnecessary margins, e.g., to transmit more capacity for a given distance or increasing the distance for a given capacity, comes down to reducing the cost per transmitted bit [1] and is therefore interesting for system vendors.

It has been shown in [2,3] that one way to reduce margins can be achieved through probabilistic design to minimize the impact of system parameter variations on transmission performance. Besides, flexible provisioning based on statistical system availability instead of fixed margins has been proposed in [4,5]. In this work we propose an *a priori* calculation of performance margins, given that input physical parameters can

be measured or monitored. A simple and computationally efficient calculation of margins, without compromising accuracy, would be valuable for resource provisioning or restoration in dynamic networks, as well as for operators that wish to upgrade their network during the system lifetime, e.g., by adding new channels, dynamically changing modulation formats, or changing the frequency allocation plan, etc.

As discussed in [6,7], margins can sum up to several decibels (dBs), and they are customarily classified into three categories: (1) “unallocated” margins that are not used because they are either unnecessary for the requested service or impossible with the installed transponder; (2) “design” margins that are used as a remedy against either the intrinsic inaccuracy of the QoT estimation tool or the limited knowledge of the network at the design stage leading to parameter uncertainties [8]; and (3) “system” margins that account for fast time-varying impairments such as polarization-dependent loss (PDL), slow time-varying impairments that include component aging and progressive channel count increase, and, finally, an additional security margin imposed by operators.

For modern dispersion unmanaged (DU) systems and inside the validity domain of the Gaussian noise (GN) model [9], SNR before FEC is a sufficient statistic to describe the

system BER. Furthermore, the accuracy of the performance prediction tool itself, i.e., the GN model or some other tool, can be assessed through exhaustive numerical simulations or experiments, and the resulting accuracy can be explicitly used as an input in the margin estimation process. Furthermore, in the context of dynamically adapting networks, several new “designs” can be made during the system lifetime, for instance, always adapting to the true aging conditions of a system in terms of losses, channel load at a given time, etc., given that this information is available to the network controller. For such networks, margins may be reclassified into “stochastic” margins, “deterministic” margins, and “security” margins, according to the nature of effects they are used against. “Stochastic” margins [10] provide resilience against SNR fluctuations, only due to fast time-varying stochastic effects like PDL or polarization mode dispersion (PMD) [11,12]. A perfect knowledge of the physical parameters of the system, e.g., characterization of the PDL elements with infinite accuracy, will enable an accurate description of the SNR fluctuations and just enough dimensioning of stochastic margins. However, if stochastic effects are not negligible, such margins can never be suppressed. On the other hand, “deterministic” margins [13] reflect the accuracy of the QoT estimation tool and the knowledge of physical parameters that can only vary slowly in time, but in all cases, they affect the SNR in a deterministic way *at some specific moment* during the system lifetime. For instance, at the system’s beginning of life (BoL), deterministic margins may reflect only coarse information of the input parameters. If, however, the true fiber losses of a cut and re-spliced fiber or the fluctuations of an amplifier noise factor (NF) are monitored, new designs with “just enough” margins can be proposed to the operator for the next design update. Having all the necessary information, the operator may then decide to trade these margins for higher capacity or higher distance, or they might maintain the unallocated margins, e.g., to counter-balance anticipated future degradation. Finally, additional “security” margins [14] can be used against effects that are not modeled by the previous framework, e.g., environmental factors, electrical power fluctuations, etc. This last kind of margin should be completely independent of input parameter values, their associated uncertainties, or the QoT estimation tool accuracy.

In this paper, we focus on deterministic margins, and we therefore consider that performance fluctuations due to stochastic effects are negligible [15]. We consider a generic DU system with no specific design constraints, and we use uncertainty propagation with correlated inputs to calculate the performance uncertainty, finally linking this uncertainty to margins. We stress the fact that the strategy of monitoring physical input parameter uncertainties and linking them to

the output performance uncertainty via a QoT tool is fundamentally different from the strategy where only the overall performance (e.g., the BER) is monitored for each link. While in the latter case it is impossible to trace back the contribution of a given parameter to the overall performance uncertainty, in the former there is a clear link between the two. The advantage of this method is that the new network state after a specific operator action, e.g., adding channels, changing the frequency allocation of channels, changing the fiber input power, etc., can be *a priori* determined, and margins can be safely reduced since there is now a guarantee that such an action does not compromise the network state.

In Section 2 we consider a simplified GN model expression for the SNR, and we derive the SNR uncertainty as a function of the uncertainties of input parameters, as similarly done in the context of 10 Gbps systems in [16,17]. A basic feature of our approach is to consider that input parameters are correlated random variables (RVs) in the most general case, with statistics that are known and invariant in the time lapse between two consecutive designs along the system lifetime. We then propose a way to set the deterministic margins based on the estimated SNR uncertainty, noting that our proposed fast computation can be used instead of cumbersome Monte Carlo assessments of the actual SNR probability density function (PDF), and it is therefore implementable in a controller. We also point out the fact that the method described in [6,7] to choose margins is equivalent to considering that all model inputs are fully correlated, resulting in a systematic overestimation of margins.

In Section 3, we assess the accuracy of our proposed method against Monte Carlo estimations of the SNR probability density function, and we indicate the validity limits of our approach. We also investigate the differences between three scenarios of interest where input parameters are uncorrelated, fully correlated, or partially correlated, and we discuss the implications in terms of margins in different propagation regimes, i.e., the linear regime, nonlinear regime, and close to the optimal power.

Finally, in Section 4 we use the developed formalism to provide some numerical illustrations that attempt to address the question of the relative impact of model inputs on performance variability.

2. PERFORMANCE MODEL

The considered dispersion unmanaged optical transmission system is shown schematically in Fig. 1. N_{ch} lasers with central wavelengths $\lambda_1, \dots, \lambda_{N_{\text{ch}}}$ are fed to a polarization division multiplexing (PDM) I/Q modulator and are then multiplexed in wavelength before being injected in the line. At the receiver end, the central channel is selected by a matched filter, and it

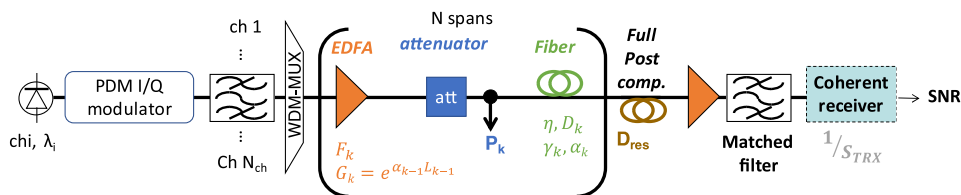


Fig. 1. Optical transmission system considered in this work.

is decoded by a coherent receiver. For this system we consider that the loss of span $(k - 1)$ is exactly recovered by the erbium-doped fiber amplifier (EDFA) of span k and that an additional attenuator is used at the input of every span as an independent “knob” for fixing the power injected in the fiber of span k . At the receiver side, we neglect the noise added by the final EDFA for simplicity, while we consider ideal dispersion compensation and matched filtering.

According to [18], for DU systems the noise-to-signal-ratio (NSR), denoted S^{-1} , is equal to the sum of the NSRs of the amplified spontaneous emission (ASE) noise, nonlinear noise [19], and transponder imperfection noise:

$$S^{-1} = S_{\text{ASE}}^{-1} + S_{\text{NL}}^{-1} + S_{\text{TRX}}^{-1}. \quad (1)$$

The transponder imperfection NSR, S_{TRX}^{-1} , can be determined through back-to-back offline experiments for a given transmitter/receiver couple. The ASE NSR, S_{ASE}^{-1} , is given by [20]

$$S_{\text{ASE}}^{-1} = \sum_{k=1}^N h c \lambda^{-1} B F_k e^{\alpha_k L_k} P_k^{-1}, \quad (2)$$

where h is the Planck constant, c is the speed of light in a vacuum, λ is the wavelength, B is the signal bandwidth, N is the number of spans, and all the other symbols are explained in Table 1. Finally, for the nonlinear noise NSR, S_{NL}^{-1} , we assume the approximation {[21], Eq. (57)}

$$S_{\text{NL}}^{-1} = \sum_{k=1}^N \eta \cdot \gamma_k^2 |D_k|^{-1} \alpha_k^{-1} P_k^2, \quad (3)$$

where we further used the approximation for long spans $L_{\text{eff}} = \frac{1-e^{-\alpha L}}{\alpha} \approx 1/\alpha$ for the effective length L_{eff} , and η is a normalized nonlinear coefficient. In this work we neglect the impact of optical filtering along the line and of linear crosstalk due to a non-infinite blocking ratio in an optical node. Furthermore, we assume all parameters to be wavelength independent and amplifier gain to be flat. As is evident from Eqs. (1)–(3), the system SNR can be written as a function of W input parameters r_l , $l = 1, \dots, W$, i.e.,

$$S = f(r_1, r_2, \dots, r_W). \quad (4)$$

In this work we consider that input parameter values are uncertain, i.e., r_l are random variables with expected values $\mu_{r_l} = E[r_l]$, while the SNR S is a multidimensional RV. As a

first step in evaluating the moment of S , we define the quantity “nominal SNR” as

$$S_{\text{nom}} = f(\mu_{r_1}, \mu_{r_2}, \dots, \mu_{r_W}), \quad (5)$$

and we denote with $S_{\text{nom,dB}}$ the same quantity in dB scale [22], i.e., $S_{\text{nom,dB}} = 10 \log_{10}(S_{\text{nom}})$. In general, $S_{\text{nom}} \neq E[S]$, since SNR is not a linear function of the input parameters r_l . However, for small variations of r_l around their average values μ_{r_l} , Eq. (4) can be linearized, and therefore $E[X] \approx E_{\text{nom}}$. Apart from the average value, in what follows we also calculate the SNR standard deviation, denoted as ΔS and referred to as SNR “uncertainty.”

We consider now a certain light path and a target capacity to deliver with a given modulation format. To determine if the path is feasible or not, we propose the following three-step procedure: (1) calculate the nominal SNR $S_{\text{nom,dB}}$; (2) calculate the SNR margin in dB (i.e., a margin relative to the SNR) M_{dB} , given the estimated input parameter uncertainties and an assessment of the intrinsic uncertainty attributed to the QoT estimator; and (3) decide that the path is feasible if $S_{\text{nom,dB}} - M_{\text{dB}} \geq S_{c,\text{dB}}$ or unfeasible if $S_{\text{nom,dB}} - M_{\text{dB}} < S_{c,\text{dB}}$. We define the critical SNR value $S_{c,\text{dB}}$ for reliable operation as

$$S_{c,\text{dB}} \triangleq (S_{\text{FEC,dB}} + M_{\text{sec,dB}}) + M_{\text{st,dB}}, \quad (6)$$

where $S_{\text{FEC,dB}}$ is the FEC limit for the considered modulation format, $M_{\text{sec,dB}}$ is the security margin, and $M_{\text{st,dB}}$ is the stochastic margin. The deterministic margin calculated in the design phase is therefore equal to

$$M_{\text{dB}} \triangleq S_{\text{nom,dB}} - S_{c,\text{dB}}. \quad (7)$$

Note that [7] takes values of margins associated with a given parameter type found in the literature and illustrates the relative importance of system, design, and unallocated margins for a given light path. However, with the formulation of [7] it is unclear how one can convert the parameter uncertainty reduction provided by in-field monitoring into additional capacity when designing system upgrades. For this, we propose to analytically set margins according to the overall uncertainty of the SNR, calculated from the uncertainties on model input parameters and the mathematically derived weights of these parameter uncertainties. To do this, we first set the deterministic margins M_{dB} as a multiple of the SNR uncertainty in dB scale ΔS_{dB} as

$$M_{\text{dB}} \triangleq n \cdot \Delta S_{\text{dB}}. \quad (8)$$

If the uncertainty ΔS is small compared to the expected value of S and $E[S]$, ΔS_{dB} can be approximated as (see Appendix A)

$$\Delta S_{\text{dB}} \approx \frac{10}{\ln 10} \frac{\Delta S}{S_{\text{nom}}}, \quad (9)$$

where we replaced $E[X] = S_{\text{nom}}$.

The choice of the value for n sets the outage or failure probability (FP), i.e., the probability that $S_{\text{nom,dB}} - M_{\text{dB}} < S_{c,\text{dB}}$. In other words, n sets the probability that the margins chosen during network design are not enough, with higher values

Table 1. Model Parameters

Symbol	Quantity
P_k	Power per channel at input of span k
F_k	Noise factor of span k amplifier
α_k	Attenuation coefficient of span k
L_k	Length of span k
γ_k	Nonlinear parameter of span k
D_k	Dispersion parameter of span k
$1/S_{\text{TRX}}$	Transponder imperfection SNR

for n designating higher margins and lower FPs. If the SNR distribution S_{dB} is Gaussian distributed, FP can be calculated analytically [23] as

$$\text{FP} = \frac{1}{2} \text{erfc} \left(\frac{n}{\sqrt{2}} \right), \quad (10)$$

where $\text{erfc}(\cdot)$ is the complementary error function. For instance, choosing $n = 1$ means that there is a 15.87% chance that the system will fail because of insufficient margins, and $n = 3$ means that the system has a 0.135% chance of failure.

Following the law of propagation of uncertainty [24], the SNR standard deviation ΔS verifies the equation [25]

$$\Delta S^2 = \sum_{l=1}^W \sum_{m=1}^W \rho_{lm} Q_{r_l} Q_{r_m}. \quad (11)$$

With ρ_{lm} we denote the correlation between r_l and r_m , and with Q_{r_l} we denote the weighted SNR uncertainty associated with the parameter r_l , defined as

$$Q_{r_l} \triangleq \left| \frac{\partial S}{\partial r_l} \right| \Delta r_l, \quad (12)$$

where Δr_l is the standard deviation or uncertainty of the input parameter r_l . Going forward, similarly to Eq. (9), the weighted SNR uncertainty of parameter r_l in dB scale is approximated as

$$Q_{r_l, \text{dB}} = \frac{10}{\ln 10} \frac{\Delta r_l}{S_{\text{nom}}}, \quad (13)$$

and the margin associated with the parameter r_l is defined as

$$M_{r_l, \text{dB}} \triangleq n \cdot Q_{r_l, \text{dB}}. \quad (14)$$

Concerning the correlation coefficient, we consider that parameters may be either fully correlated (positively or negatively) or totally uncorrelated, i.e.,

$$\rho_{lm} = \begin{cases} 1, & m = l \\ 1, & r_l \text{ and } r_m \text{ correlated} \\ 0, & r_l \text{ and } r_m \text{ uncorrelated} \end{cases}. \quad (15)$$

We now consider three cases of practical interest. When all parameters are uncorrelated, Eq. (11) simplifies to the quadratic sum of the weighted uncertainties Q_{r_l} , i.e.,

$$\Delta S_U = \sqrt{\sum_{l=1}^W Q_{r_l}^2}. \quad (16)$$

Combining Eqs. (8), (9), (14), and (16), we can easily show that the corresponding margin is equal to

$$M_{U, \text{dB}} = \sqrt{\sum_{l=1}^W M_{r_l, \text{dB}}^2}. \quad (17)$$

On the other hand, when all parameters are fully correlated, Eq. (11) simplifies to the direct sum of Q_{r_l} (see Appendix B), i.e.,

$$\Delta S_F = \sum_{l=1}^W Q_{r_l}, \quad (18)$$

and the corresponding margin is given by

$$M_{F, \text{dB}} = \sum_{l=1}^W M_{r_l, \text{dB}}. \quad (19)$$

Finally, input parameters may be partially correlated when correlations take arbitrary values. In this case the more general Eqs. (11)–(15) need to be used to yield the uncertainty $\Delta S_{P, \text{dB}}$ and the margin $M_{P, \text{dB}}$. As will be discussed later, in this work we focus on the specific case of partial correlation with $\rho_{lm} = 1$ when r_l and r_m are parameters of the same type and $\rho_{lm} = 0$ when r_l and r_m are parameters of different types. For instance, power levels at the input of different spans are correlated, but powers are not correlated with amplifier NFs.

In practice, the standard deviations or uncertainties of the input parameters may be found from component datasheets or measurements, and they reflect either the industrial process followed to deliver a component or the instrument accuracy used to characterize it. Therefore, a field measurement (or monitoring) basically makes sense if the new measurement uncertainty is lower than the initial uncertainty obtained by datasheet information. Similarly, correlations could be set empirically by exploiting knowledge relative to the system components. In the context of system design, it makes sense to consider that some parameters are indeed fully correlated with $\rho_{lm} = 1$. For instance, noise factors of amplifiers coming from the same production line may incorporate a systematic bias. Another possibility is that the correlation terms can model “hidden” interdependences of two input parameters that depend both on some other external parameter, such as temperature [26–28]. On the other hand, other parameters may have no obvious reason to be correlated, e.g., amplifier noise factors and fiber nonlinear parameters. In all cases, offline laboratory characterizations and field data can be used to set parameter correlations in the design phase.

We note that, in [6,7], margins are calculated as the linear sum of parameter margins, and this is therefore equivalent to considering that summated margins are related to parameters that are fully correlated. For a given set of parameter uncertainties Δr_l , $\Delta S_F \geq \Delta S_U$, whatever the model for S . Therefore, considering the approach of [7] comes down to systematically over-dimensioning margins with respect to a more pragmatic approach.

3. ACCURACY ASSESSMENT

We now assess the accuracy of the uncertainty expressed in Eq. (11) for a typical DU system with N spans of standard single-mode fiber (SSMF), 32 Gbaud PDM quadrature phase-shift keying (QPSK) modulation, 15 channels with channel spacing $\delta F = 50$ GHz, a variable power P at the input of each span, and all other parameters having identical uncertainties and nominal values for all spans that appear in Table 2 if not mentioned otherwise. We analyze the central channel, for which the normalized nonlinear parameter has been assessed to be $\eta = 299.49$ ps/nm as a best fit from numerical simulations [29,30] and is assumed to be known without uncertainty [31]. The SNR statistics are then obtained by randomly drawing the

Table 2. Numerical Values Used in Our Example^a

Quantity	Mean Value	Uncertainty
$P_{dB,k}$	0 dBm	1 dB
$F_{dB,k}$	6 dB	0.5 dB
α_k	0.22 dB/km	0.01 dB/km
L_k	100 km	1 km
D_k	16.7 ps/nm/km	0.2 ps/nm/km
γ_k	1.317 W ⁻¹ km ⁻¹	0.1317 W ⁻¹ km ⁻¹
$S_{TRX,dB}$	22.5 dB	2.5 dB

^a All parameters are assumed to admit uncertainties that are expressed in the same scale as their mean values, e.g., uncertainty for powers and NFs are expressed in dB (denoted with the index “dB”), uncertainty for attenuation coefficients in decibels/kilometer (dB/km), etc. The numerical values that are considered in this example are indicative and do not represent the general case. Different fabrication processes and/or different instrument accuracies should yield different uncertainties.

parameter sets $N_{MC} = 10^5$ times, using when needed the conversion between linear and dB scale uncertainties as detailed in the previous section.

In Fig. 2, we plot the nominal value of SNR S (left axis) and the SNR uncertainty ΔS (right axis), as a function of the input power P , separately for the three correlation scenarios discussed in the previous section, i.e., (a) uncorrelated inputs, dubbed “ucorr”; (b) partially correlated inputs, where parameters of the same type are fully correlated and parameters of different types are uncorrelated, dubbed “pcorr”; and (c) fully correlated inputs, dubbed “fcorr” [32]. Markers denote Monte Carlo (MC); the solid line indicates the SNR prediction of Eq. (1); and dashed, dotted, and dashed-dotted lines indicate the prediction of Eq. (11) (theory). First, we note that in all cases, ΔS is well approximated by Eqs. (9) and (11) with an error always lower than 0.05 dB. Second, commenting on the difference between the three correlation scenarios in terms of the calculated uncertainty, we note that, for full correlation, uncertainties reach up to about 2 dB more with respect to the uncorrelated case, whereas the partially correlated scenario lies somewhere between the two, with about 1 dB difference compared to full correlation. Given the relation between uncertainty and margins of Eq. (8) and for, e.g., $n = 2$, we can conclude that, for this specific configuration, we would need 4 dB more margins for the fully correlated case compared to the uncorrelated case. Finally, we point out that in the proximity of the optimal power, SNR uncertainty appears to be minimized for “pcorr” and “fcorr.” This can be qualitatively understood by the fact that the derivative of the SNR with respect to power [a quantity that multiplies the power uncertainty in Eq. (11)] is close to zero for this operation regime and the margin associated with power is also minimized. However, in the “ucorr” case, the contribution of power uncertainty to the overall uncertainty is clearly not dominant as in the previous two scenarios, and therefore no clear minimization is observed.

In Fig. 3 we plot the nominal SNR (left axis, solid line) and the uncertainty ΔS (right axis) as a function of the number of spans N , with markers denoting Monte Carlo and lines denoting predictions, for the three correlation scenarios as before. As in the previous example, the overall SNR uncertainty predicted by Eq. (11) is in very good agreement with uncertainty

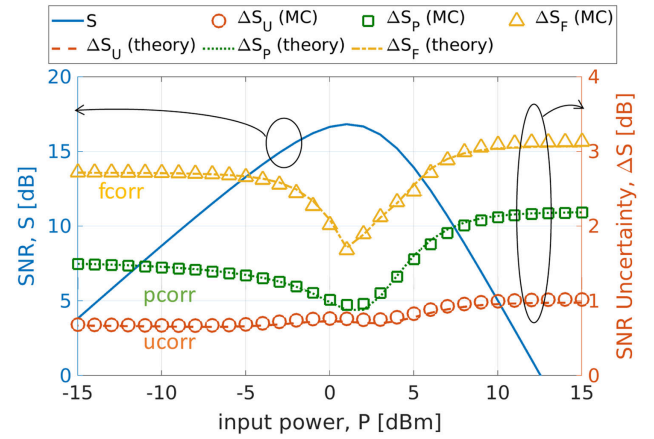


Fig. 2. Nominal SNR versus input power P (left axis) and uncertainty ΔS (right axis) by Monte Carlo (markers) and Eq. (11) (lines) for the three considered scenarios: (a) no correlation between input parameters, named “ucorr”; (b) partial correlation where input parameters of the same type are fully correlated and parameters of different types are uncorrelated, named “pcorr”; and (c) full correlation between input parameters, named “fcorr.”

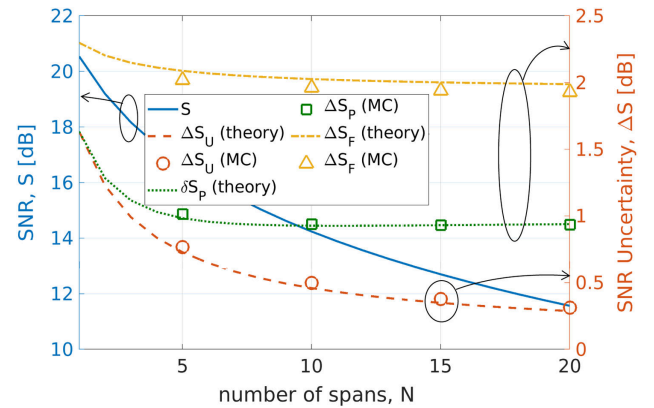


Fig. 3. Nominal SNR for $P = 0$ dBm versus number of spans N (left axis) and SNR uncertainty (right axis) from theory (lines) and Monte Carlo (markers).

obtained by Monte Carlo. Furthermore, we note that uncertainty decreases continuously with the number of spans for the “ucorr” scenario while it quickly saturates for the “fcorr” and “pcorr” scenarios after $N \approx 5$ spans. For “fcorr” this can be qualitatively understood from Eq. (18) and supposing that there is only one uncertain parameter, e.g., power. Reasoning in terms of the NSR, S^{-1} , we note that both S^{-1} and its uncertainty are proportional to the number of spans N . Following then Eq. (A5) of Appendix A we can simply note that the uncertainty in dB saturates to ΔP in the linear regime and to $2\Delta P$ in the nonlinear regime, and therefore suggests that margins can be roughly set to be independent of the number of spans. At the other extreme, for “ucorr,” the uncertainty is proportional to $\sqrt{N}$ due to the quadratic sum, appearing, e.g., in Eq. (16), and therefore the corresponding asymptotes are $\Delta P/\sqrt{N}$ and $2\Delta P/\sqrt{N}$. Similar asymptotes can be derived for all parameters, resulting in an overall “ucorr” uncertainty in

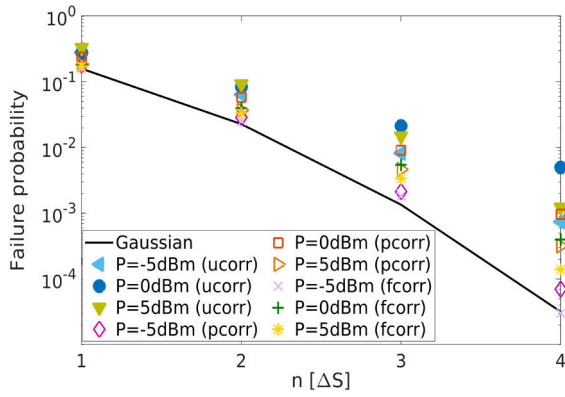


Fig. 4. Failure probability versus n for a system with $N=5$ spans and three different power levels, i.e., -5 , 0 , and 5 dBm per channel. Monte Carlo simulations are indicated with different marker types for the “ucorr,” the “pcorr,” and the “fcorr” scenario. We note that ΔS is calculated differently for each scenario of correlation and power, logically leading to different values of calculated margins.

dB that decreases with the number of spans as shown in Fig. 3. Figure 3 therefore suggests that, when input parameters are uncorrelated, margins can be set in a distance-specific way, i.e., allocating higher margins to short links compared to the lower margins needed for longer links.

In order to illustrate the impact on margins, in Fig. 4 we plot the failure probability as a function of parameter n used to set margins as described in Eq. (8). With a solid line we plot the Gaussian reference of Eq. (10), and with markers we denote the failure probability obtained by Monte Carlo for the three correlation scenarios “ucorr,” “pcorr,” and “fcorr,” as well as for three different power levels in the linear regime ($P = -5$ dBm), nonlinear regime ($P = 5$ dBm), and close to the optimal power ($P = 0$ dBm). We note that even though the SNR standard deviation is fairly well approximated by Eq. (11) as discussed previously in this section, there is a non-negligible difference between the predicted failure probability and the one obtained by Monte Carlo. This can be understood by the difference between the true SNR PDF and the Gaussian PDF with parameters computed as before.

To further clarify the previous point, in Fig. 5 we plot the probability density function of the SNR obtained by Monte Carlo for the three correlation scenarios and $P = 0$ dBm. In all cases, solid lines represent a Gaussian distribution with average $S_{\text{nom,dB}}$ and standard deviation ΔS_{dB} calculated by Eqs. (1) and (11), while dashed lines indicate the $\pm 3\Delta S_{\text{dB}}$ reference, with the arrow indicating the suggested margin for $n = 3$. As is obvious, the Gaussian approximation (solid line) does not always capture the fine details of the PDF such as the mean SNR (as opposed to the SNR nominal value) or the PDF tails, which are not Gaussian in the general case. We suggest that such a mismatch is due to the first-order approximations employed in the paper. Nevertheless, we note that Eq. (11) (which is evaluated analytically) seems to provide a fair estimate of the SNR standard deviation that takes into account parameter correlation, while the accuracy of the estimation is expected to improve for lower parameter uncertainties.

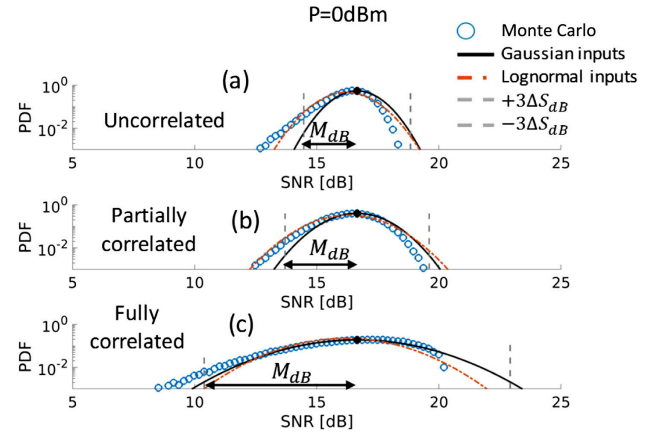


Fig. 5. SNR PDFs evaluated with Monte Carlo for a system with $N=5$ spans, all other parameters appearing in Table 2, and for the three considered correlation scenarios: (a) uncorrelated RVs, (b) partially correlated RVs, and (c) fully correlated RVs. Lines represent Gaussian distributions with average $S_{\text{nom,dB}}$ and standard deviation ΔS_{dB} calculated by Eqs. (1) and (11), with Gaussian input RVs (solid lines) and lognormal input RVs (dashed-dotted lines). Dashed lines denote the $\pm 3\Delta S_{\text{dB}}$ limits, and arrows denote the corresponding margin $M_{\text{dB}} = n \cdot \Delta S_{\text{dB}}$, with $n = 3$.

An improvement of the prediction provided by Eqs. (1) and (11) can be obtained by supposing that (a) each elementary NSR contribution of Eq. (1) is independent and (b) all NSR random variables are lognormally distributed, something that can be obtained if we neglect the uncertainty of the fiber length for the linear noise part and uncertainty of the dispersion and attenuation in the nonlinear noise part. Given the above assumptions, the overall NSR PDF can be approximated by a new lognormal PDF in linear scale, i.e., a new normal distribution is dB scale, with parameters calculated as in [33]. This approximation is accurate, especially for the left tail of the SNR PDF, as shown in Fig. 5 (dashed-dotted line), i.e., the tail that is mainly involved in the FP calculation. However, we note that, while considering lognormal RVs offers a better approximation for the “ucorr” and “pcorr” PDFs, we observe a slight degradation for “fcorr.” In Fig. 6 we report the same results of failure probability as in Fig. 4, where the mean S_{dB} and standard deviation ΔS_{dB} of the SNR have been modified according to [33]. Comparing Figs. 6 and 4 we observe that, while accuracy degrades for high values of n , a very good match is observed between the reference formula and Monte Carlo for $n = 1$ at the expense, however, of a more complicated approach and additional calculations. In all cases this means that, for example, maintaining the same failure probability with different correlation scenarios would necessitate about 1–2 dB of extra margins for the fully correlated scenario compared to an uncorrelated scenario.

4. RELATIVE IMPACT OF PARAMETERS ON UNCERTAINTY

We now focus on the relative impact of the input parameter uncertainties on SNR uncertainty ΔS as a function of each one of the input parameter uncertainties ΔX as these are

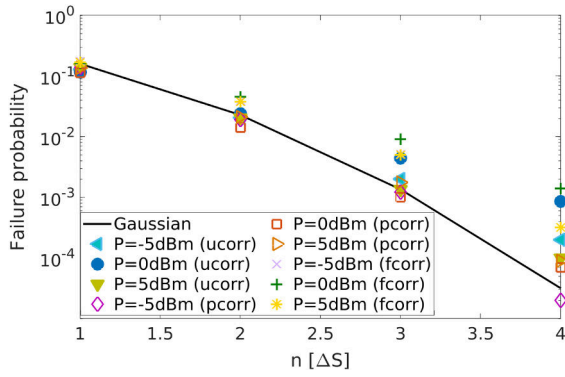


Fig. 6. Failure probability versus n for a system with $N=5$ spans and three different power levels, i.e., -5 , 0 , and 5 dBm per channel, using the lognormal left tail approximation of [33].

expressed in Table 2. Pointing out the order of importance of physical parameters with respect to their impact on the overall uncertainty can help us point out which parameters most merit being regularly monitored. An answer can be obtained by the weighted uncertainty of each parameter of Eq. (12), which involves the derivative with respect to the parameter r_l and therefore expresses the sensitivity of the SNR with respect to variations of this parameter.

In Fig. 7 on the right axis we plot the quantities Q_{r_l} by setting $\Delta r_l = 1$ to perceive their relative impact independently of the main parameter uncertainty. As can be seen from Eq. (12), this is equivalent to plotting the quantity $|\partial S / \partial r_l|$, which is also called the *sensitivity coefficient* of the parameter r_l [24], as it reflects the sensitivity of r_l to uncertainty, as this is perceived on the overall SNR uncertainty. The SNR uncertainties ΔS are estimated using Eq. (12) as a function of the input power P for a system with $N=5$ spans, separately for all parameters, when all other uncertainties are set to zero and all other system parameters are set according to Table 2. On the left axis we plot the corresponding SNR for reference, and on the right axis we plot the SNR uncertainty ΔS [dB] in logarithmic scale to distinguish contributions of different orders of magnitude. We note that even though the exact order of importance strongly depends on the input power, the fiber attenuation is clearly dominant in all propagation regimes for the investigated system parameters. Close to the optimal power, the nonlinear parameter γ is also critical, followed by the transponder imperfection noise, the input power, the noise factor, etc. We note that, in this case, no or partial correlation will not qualitatively change the order of importance but will only change the absolute values by a factor of $\sqrt{N}$. Furthermore, extending the findings of [2], we see that the channel power can also be used to control the impact of various parameters on the SNR uncertainty. For instance, if uncertainty on α is high while uncertainty on P and γ is low, operating slightly in the nonlinear regime is preferable to reduce margins. However, if uncertainty on the nonlinear coefficient is disproportionately high compared to the other uncertainties, operating in the linear regime could present an advantage with respect to the overall uncertainty.

Finally, in Fig. 8 we illustrate the parameter uncertainties as before, this time as a function of the number of spans N

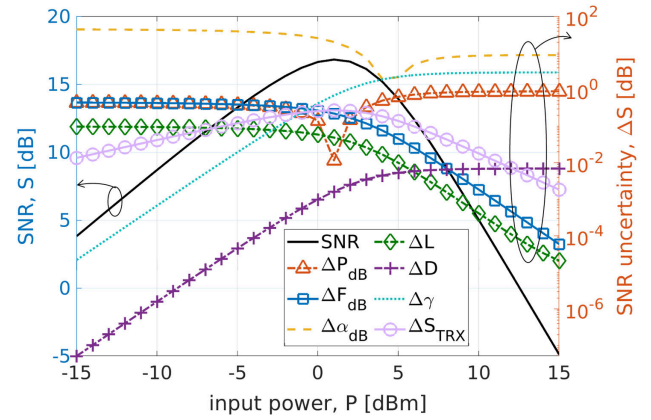


Fig. 7. SNR versus input power (left axis) and separate uncertainties (right axis) of power, noise figure, attenuation, span length, dispersion parameter, nonlinear parameter, and transponder NSR.

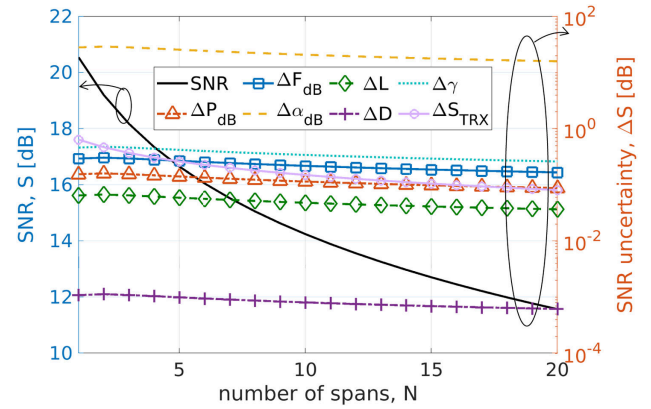


Fig. 8. SNR versus number of spans N (left axis) for a power of $P=0$ dBm, and separate SNR uncertainties versus N (left axis) and separate uncertainties (right axis) of power, noise figure, attenuation, span length, dispersion parameter, nonlinear parameter, and transponder imperfection noise.

and for a fixed power $P=0$ dBm. We note that the main tendencies concerning the importance of different parameters are maintained for all quantities. We also note that while, for instance, the transponder NSR is the second most important quantity after fiber attenuation, its importance drops significantly over distance for the considered system configuration and the numerical values used in this example.

5. CONCLUSION

In the context of dispersion unmanaged transmission with dedicated monitoring of the physical system parameters, we employ uncertainty propagation to connect the overall performance uncertainty to the input parameter uncertainties and to accordingly set margins. We numerically establish the accuracy of our approach for a typical system, and we illustrate the consequences in terms of margins for different correlation scenarios ranging from uncorrelated inputs to fully correlated inputs. We also propose an additional correction to the classical

uncertainty propagation that improves the accuracy of predicting the system failure probability by better capturing the probability distribution left tail and mean value. Our results indicate that for a typical system, 1–2 dB on SNR margins can be saved if parameters are truly uncorrelated, compared to legacy methods that are equivalent to considering full correlation between all input parameters. We finally point out that such a framework can be used to classify parameters with respect to their relative contribution to the overall uncertainty, therefore indicating which ones most merit being monitored. Alternatively, the same framework can be used to (re)design systems, e.g., in terms of fiber injection powers, in a way that minimizes performance uncertainty according to the available input parameter uncertainties, resulting in a minimization of performance margins. Including the effects of optical filtering and extending the analysis to stochastic margins is left for future investigation.

APPENDIX A

Given a random variable X with a positive expected value $E[X] = X_0$, X in dB scale can be approximated by a first-order Taylor expansion around X_0 as

$$Y = 10\log_{10} X \approx \frac{10}{\ln 10} \left[\ln(X_0) + \frac{X}{X_0} - 1 \right]. \quad (\text{A1})$$

The variance of Y can be evaluated as [23]

$$\text{Var}[Y] = E[Y^2] - E[Y]^2. \quad (\text{A2})$$

Plugging Eq. (A1) into (A2) and after some elementary algebra, we can evaluate the variance of Y as $\text{Var}[Y] = \left(\frac{10}{\ln 10}\right)^2 \frac{\text{Var}[X]}{E[X]^2}$, and therefore the standard deviation of Y , $\Delta Y = \sqrt{\text{Var}[Y]}$ can be approximated as

$$\Delta Y \approx \frac{10}{\ln 10} \frac{\Delta X}{X_0}. \quad (\text{A3})$$

In other words, if the parameter r_l is associated with an uncertainty Δr_l that is small with respect to the expected value of r_l , then in dB scale the quantity $(r_l \pm \Delta r_l)$ reads

$$(r_l \pm \Delta r_l)_{\text{dB}} \approx r_{l,\text{dB}} \pm \Delta r_{l,\text{dB}}, \quad (\text{A4})$$

where the parameter uncertainty in dB is approximated as

$$\Delta r_{l,\text{dB}} \approx \left| \frac{\partial(10\log_{10} r_l)}{\partial r_l} \right| \Delta r_l = \frac{10}{\ln 10} \frac{\Delta r_l}{r_l}. \quad (\text{A5})$$

Therefore, uncertainties in linear scale can be converted (in the first order) into uncertainties in dB scale using Eqs. (A4) and (A5).

APPENDIX B

For fully correlated inputs, Eq. (11) can be written as

$$\Delta S^2 = \sum_{l=1}^W \sum_{m=1}^W Q_{r_l} Q_{r_m} = \sum_{l=1}^W Q_{r_l}^2 + 2 \sum_{l=1}^W \sum_{m=1}^{l-1} Q_{r_l} Q_{r_m} = \left(\sum_{l=1}^W Q_{r_l} \right)^2 = \Delta S_F^2, \text{ with } \Delta S_F = \sum_{l=1}^W Q_{r_l} \text{ as defined by Eq. (18).}$$

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REFERENCES AND NOTES

1. J. Pesic, T. Zami, P. Ramantanis, and S. Bigo, "Faster return of investment in WDM networks when elastic transponders dynamically fit ageing of link margins," in *Optical Fiber Communication Conference* (Optical Society of America, 2016), paper M3K.2.
2. S. J. Savory and H. Chin, "Probabilistic design of nonlinear optical transmission systems," in *Optical Fiber Communications Conference and Exhibition (OFC)*, March 2016, pp. 1–3.
3. H. M. Chin, D. Charlton, A. Borowiec, M. Reimer, C. Laperle, M. O'Sullivan, and S. J. Savory, "Probabilistic design of optical transmission systems," *J. Lightwave Technol.* **35**, 931–940 (2017).
4. I. Kim, X. Wang, M. Bouda, O. Vassilieva, Q. Zhang, P. Palacharla, and T. Ikeuchi, "Q-availability based virtual optical network provisioning," in *Optical Fiber Communications Conference and Exposition (OFC)*, March 2018, pp. 1–3.
5. I. Kim, X. Wang, O. Vassilieva, P. Palacharla, and T. Ikeuchi, "Maximizing optical network capacity through SNR-availability based provisioning," in *Optical Fiber Communication Conference (OFC)* (Optical Society of America, 2019), paper M4J.1.
6. Y. Pointurier, "Design of low-margin optical networks," *J. Opt. Commun. Netw.* **9**, A9–A17 (2017).
7. J. L. Augé, "Can we use flexible transponders to reduce margins?" in *Optical Fiber Communication Conference and Exposition and the National Fiber Optic Engineers Conference (OFC/NFOEC)* (2013), pp. 1–3.
8. Joint Committee for Guides in Metrology, "Evaluation of measurement data—Supplement 1 to the "Guide to the expression of uncertainty in measurement,"—Propagation of distributions using a Monte Carlo method," JCGM 101:2008 (Bureau International des Poids et Mesures, 2008).
9. P. Poggiolini, "The GN model of non-linear propagation in uncompensated coherent optical systems," *J. Lightwave Technol.* **30**, 3857–3879 (2012).
10. Previously part of "system" margins.
11. N. Rossi, A. Ghazisaeidi, and P. Ramantanis, "Stochastic nonlinear interference in dispersion coherent optical links," in *42nd European Conference on Optical Communications (ECOC)* (2016).
12. P. Serena, S. Musetti, S. Almonacil, S. Bigo, A. Benoni, P. Jenneve, N. Rossi, and P. Ramantanis, "The Gaussian noise model extended to polarization dependent loss and its application to outage probability estimation," in *European Conference on Optical Communication (ECOC)*, September 2018, pp. 1–3.
13. Previously part of "design", "system" and "unallocated" margins.
14. Previously part of "system" margins.
15. Other transient effects and power instability are also overlooked.
16. D. Penninckx and C. Perret, "New physical analysis of 10-Gb/s transparent optical networks," *IEEE Photon. Technol. Lett.* **15**, 778–780 (2003).
17. T. Zami, A. Morea, and F. Leplingard, "A new method to plan more realistic optical transparent networks," *Bell Labs Tech. J.* **14**, 213–226 (2010).
18. F. Vacondio, O. Rival, C. Simonneau, E. Grellier, A. Bononi, L. Lorcy, J.-C. Antona, and S. Bigo, "On nonlinear distortions of highly dispersive optical coherent systems," *Opt. Express* **20**, 1022–1032 (2012).
19. Signal depletion (or "droop") is also neglected for simplicity.
20. G. P. Agrawal, *Fiber-Optic Communication Systems*, 4th ed. (Wiley, 2010).
21. P. Poggiolini, G. Bosco, A. Carena, V. Curri, Y. Jiang, and F. Forghieri, "The GN-model of fiber non-linear propagation and its applications," *J. Lightwave Technol.* **32**, 694–721 (2014).

22. In what follows, we denote with X_{dB} (or $X_{i,\text{dB}}$) the quantity $10\log_{10}X$ [or $10\log_{10}(X_i)$].
23. A. Papoulis, *Probability, Random Variables, and Stochastic Processes*, 3rd ed. (McGraw-Hill, 1991).
24. B. N. Taylor and C. E. Kuyatt, *Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results* (1994).
25. With respect to Eq. (A-3) of [18] we replaced $u(x_i, x_j) = \rho_{ij}u(x_i)u(x_j)$, where ρ_{ij} is the Pearson correlation coefficient, and we rearranged the terms to get $(\partial f/\partial x_i)(\partial f/\partial x_j)u(x_i, x_j) = (\partial f/\partial x_i)(\partial f/\partial x_j)\rho_{ij}u(x_i)u(x_j) = \rho_{ij}(\partial f/\partial x_i)u(x_i)(\partial f/\partial x_j)u(x_j) = \rho_{ij}Q_iQ_j$.
26. J. H. Lee, W. J. Lee, and N. Park, "Comparative study on temperature-dependent multichannel gain and noise figure distortion for 1.48- and 0.98- μm pumped EDFAs," *IEEE Photon. Technol. Lett.* **10**, 1721–1723 (1998).
27. F. A. Flood, "Comparison of temperature dependence in C-band and L-band EDFAs," *J. Lightwave Technol.* **19**, 527–535 (2001).
28. C. Berkdemir and S. Ozsoy, "On the temperature-dependent gain and noise figure analysis of C-band high-concentration EDFAs with the effect of cooperative upconversion," *J. Lightwave Technol.* **27**, 1122–1127 (2009).
29. Raman scattering has been neglected in split step Fourier simulations.
30. A. Bononi, N. Rossi, and P. Serena, "On the nonlinear threshold versus distance in long-haul highly-dispersive coherent systems," *Opt. Express* **20**, B204–B216 (2012).
31. If an accuracy assessment of the QoT tool is available, this can be used as an input of the η parameter uncertainty.
32. Since variation of parameters does not necessarily imply variation of the SNR in the same way, for the "fcorr" scenario we chose parameters to be either positively or negatively correlated ($\rho = \pm 1$) in such a way that always leads to a lower SNR; i.e., fully correlated inputs correspond to a "worst case."
33. S. Asmussen and L. Rojas-Nandayapa, "Asymptotics of sums of lognormal random variables with Gaussian copula," *Stat. Probab. Lett.* **78**, 2709–2714 (2008).